FL-PFA: A Polar Format Algorithm for Wide-Beam Forward-Looking SAR Imaging Integrating Spatial-Variant Motion Compensation

Zhichao Meng, Lei Zhang, *Member, IEEE*, Jingyue Lu, and Xinshuo Wang

***Abstract*— Multichannel forward-looking synthetic aperture radar (FLSAR) imaging presents challenges due to spatiotemporal coupling, especially in wide-beam scenarios. This article introduces a new polar format algorithm (PFA) called the forward-looking polar format algorithm (FL-PFA) for wide-beam FLSAR imaging. The proposed algorithm utilizes a time–space–time hybrid scheme to address the spatiotemporal coupling, matching the spatiotemporal structure of FLSAR echoes. Furthermore, the scheme combines spatial-variant motion compensation (MOCO) and polar format imaging, resulting in an efficient processing flow. For spatial-variant MOCO, this article introduces an adaptive subaperture topography- and aperture-dependent (ASATA) algorithm, offering the advantage of balancing compensation accuracy and efficiency using adaptive optimal subapertures. Subsequently, a novel PFA is proposed to obtain a focused FLSAR image. In the PFA, the azimuth angle wavenumber considers both the spatial-variant quadratic phase and residual motion errors, aligning with the spatial-variant Doppler modulation rate characteristic of FLSAR. By resampling the azimuth angle wavenumber, a well-focused FLSAR image on the polar coordinate grid and more accurate MOCO can be achieved simultaneously. Finally, extensive experiments using simulated and actual synthetic aperture radar (SAR) data demonstrate the superiority of FL-PFA.**

***Index Terms*— Adaptive subaperture topography- and aperture-dependent (ASATA), forward-looking synthetic aperture radar (FLSAR), polar format algorithm (PFA), spatial-variant motion compensation (MOCO).**

I. Introduction

FORWARD-LOOKING synthetic aperture radars (FLSARs) have gained popularity in recent years because of their exceptional mapping capabilities for the area in front of the platform [1], [2], [3]. They are widely used for remote sensing mapping, battlefield surveillance, weapon guidance, and other applications. However, FLSAR signal processing faces three significant challenges due to its unique imaging geometry configuration, which sets it apart from traditional side-looking synthetic aperture radar (SAR) systems. These challenges include left–right Doppler aliasing, spatial-variant Doppler modulation rate, and motion error with left–right anisotropy [4], [5]. The first challenge arises from symmetrical scatterers positioned on both sides of the trajectory, resulting in identical phase histories and leading to left–right Doppler aliasing. The second challenge involves the spatial variation of the Doppler frequency modulation rate, creating complications for azimuth match filtering. The last and trickiest challenge is the presence of motion error left–right anisotropy. In the left and right scenes, the phase error resulting from cross-track motion error exhibits an opposite sign and spatial variation. These critical issues significantly impede the progress of wide-beam FLSAR imaging signal processing and are known as the spatiotemporal coupling characteristics of FLSAR, especially prominent in the wide-beam scenario.

Researchers have developed two alternative systems for FLSAR imaging: bistatic FLSAR [6], [7] and multichannel FLSAR [8], [9]. The bistatic FLSAR system utilizes two platforms to overcome the undefined zone constraint of traditional SAR imaging. Typically, this system consists of either a side-looking transmitter and a forward-looking receiver on separate moving platforms or a stationary transmitter and a moving receiver [10], [11]. The imaging geometry of bistatic FLSAR is altered, resulting in a phase history of echoes equivalent to a monostatic squint-looking SAR system. This eliminates issues such as left–right Doppler aliasing and motion error left–right anisotropy. Consequently, algorithms such as range migration algorithm (RMA), chirp scaling algorithm (CSA), and PFA [12], [13], [14], which are used for monostatic squint-looking SAR, can be applied for processing bistatic FLSAR data. However, equipment and system design complexity limit the widespread use of bistatic FLSAR. In contrast, the multichannel FLSAR system achieves forward imaging by using a transmitting channel and multiple receiving channels mounted on the same platform, making it a more practical and efficient solution. In recent years, there has been a growing interest in the multichannel FLSAR system, leading to the development of numerous imaging methods [15], [16], [17], [18].

For multichannel FLSAR, the monopulse angle estimation algorithm is a simple and reliable method [19], [20], [21]. It can distinguish scatters from different azimuth angles using angle estimation algorithms, such as digital beam forming (DBF) and the sum–difference direction of arrival estimation (SD-DOA). In addition, super-resolution angle estimation

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algorithms based on sparse recovery have also been applied to generate high-resolution FLSAR imagery [22], [23]. However, these algorithms have limitations. On one hand, their precision in estimating angles is limited by a low signal-to-noise ratio (SNR) as they involve only a few pulses. On the other hand, they can achieve only limited azimuth resolution as they rely solely on real-aperture and disregard the potential of synthetic aperture. Lu et al. [4] developed a new multichannel FLSAR imaging algorithm based on the spectrum analysis (SPECAN) algorithm to achieve higher azimuth resolution. This algorithm first generates the collapsed, focused SAR image using azimuth deramp. Then, it solves the left–right Doppler aliasing by utilizing adaptive beam formation (ABF) [24], which ultimately produces high-resolution FLSAR images without ambiguity. However, the Doppler spread restricts the improvement in resolution due to the residual azimuth-variant quadratic phase when the synthetic aperture is large or the azimuth beam is wide. Furthermore, the fast decomposition back projection (FFBP) algorithm [25] is capable of generating high-resolution FLSAR images. Point-to-point accumulation allows FFBP to overcome Doppler spread and yield nonambiguous FLSAR imagery with improved azimuth resolution. However, the practical application of the FFBP algorithm is restricted due to its complexity and imprecise 2-D interpolation.

This article introduces a new polar format algorithm (PFA) for multichannel wide-beam FLSAR imaging (FL-PFA) to handle these issues. In comparison with existing FLSAR imaging algorithms, this algorithm offers the following contributions.

1) A time–space–time hybrid processing scheme is proposed to solve the spatiotemporal coupling, laying the foundation for compensating for spatial-variant motion compensation (MOCO). By incorporating the spatial-variant MOCO and azimuth focusing algorithm into this scheme, we achieve a gradual imaging process that starts with low resolution and progresses to high resolution.
2) An adaptive adaptive subaperture topography- and aperture-dependent (ASATA) algorithm is introduced to compensate for spatial-variant motion errors. This algorithm addresses the left–right anisotropy of motion errors by independently compensating for each side after the spatiotemporal decoupling process. It is worth noting that utilizing the adaptive subaperture balances efficiency and accuracy.
3) A new definition of azimuth angle wavenumber has been developed, forming the basis for a novel polar imaging algorithm. This angle wavenumber is determined by both the spatial-variant quadratic phase and residual motion errors. The definition is in line with the spatial-variant Doppler modulation rate characteristics of FLSAR, making it suitable for wide-beam FLSAR applications. Importantly, the proposed algorithm allows for the simultaneous achievement of image focusing and motion compensation (MOCO) by resampling the angle wavenumber.

The rest of this article is organized as follows. Section II describes the precise geometry and signal model for multichannel FLSAR. In Section III, the proposed algorithm is introduced. The extensive experiments with simulated and real-measured airborne SAR data are carried out in Section IV. Finally, this article is concluded in Section V.

II. Signal Model of Multichannel Forward-Look SAR Echo

This section introduces the baseband echo signal model of multichannel FLSAR in detail. Let us consider a multichannel FLSAR system with a pitch angle φ, and the data acquisition geometry is shown in Fig. 1.

First, the platform flies on the height H with a constant velocity v in the ideal condition. At a specific moment, the nominal antenna phase center (APC) is $A(x, 0, H)$, while the actual APC is $A'(x+x_e, y_e, H+z_e)$ due to atmospheric turbulence. x_e, y_e, and z_e are the three-axis deviations, respectively. Second, the uniform line array comprises M identical antenna units, and the array space is d. The middle array element is a transceiver, and the other are receivers. According to the geometry model, the actual instantaneous range $R_e(x, r_\mathrm{p}, \theta_\mathrm{p})$ from APC to the point $P_\mathrm{p}(r_\mathrm{p}, \theta_\mathrm{p})$ is represented by

$$R_e(x, r_\mathrm{p}, \theta_\mathrm{p}) = \sqrt{(x + x_e - x_\mathrm{p})^2 + (y_e - y_\mathrm{p})^2 + (H + z_e)^2} \tag{1}$$

where $x = vt$, t is the slow time, and e denoted the motion error. In addition,

$$x_\mathrm{p} = \sqrt{(r_\mathrm{p}\cos\theta_\mathrm{p})^2 - H^2} \tag{2a}$$

$$y_\mathrm{p} = r_\mathrm{p} \cdot \sin\theta_\mathrm{p}. \tag{2b}$$

Furthermore, when the SAR system transmits a linear frequency modulate (LFM) signal with centroid frequency f_c and bandwidth B, the single-channel baseband echo signal can be expressed as follows:

$$s_e^0(\tau, x, r_\mathrm{p}, \theta_\mathrm{p}) = \mathrm{rect}\left(\frac{\tau}{T_\mathrm{p}}\right) \cdot \exp\left[j\pi\gamma\left(\tau - \frac{R_e(x, r_\mathrm{p}, \theta_\mathrm{p})}{c}\right)^2\right] \cdot \exp\left(-j4\pi f_\mathrm{c} \cdot \frac{R_e(x, r_\mathrm{p}, \theta_\mathrm{p})}{c}\right) \tag{3}$$

where τ is the fast-time variable, T_p is the pulse duration, $\gamma = B/T_p$ is the modulate ratio, and c is the light speed.

Accordingly, the range wavenumber domain signal after range compressing can be given by

$$s_e^0(k_\mathrm{r}, x, r_\mathrm{p}, \theta_\mathrm{p}) = \exp\left[-jk_\mathrm{r} \cdot R_e(x, r_\mathrm{p}, \theta_\mathrm{p})\right] \tag{4}$$

where $k_\mathrm{r} = 4\pi f_\mathrm{r}/c$ is the range wavenumber, and $f_\mathrm{r} \in [-B/2, B/2] + f_\mathrm{c}$.

The Doppler modulation rate can be obtained by taking the derivative of the second exponential term in (3)

$$k_a(x, \theta_\mathrm{p}) = \frac{1}{R_e(x, r_\mathrm{p}, \theta_\mathrm{p})} - \frac{(x + x_e - x_\mathrm{p})^2}{[R_e(x, r_\mathrm{p}, \theta_\mathrm{p})]^3}. \tag{5}$$

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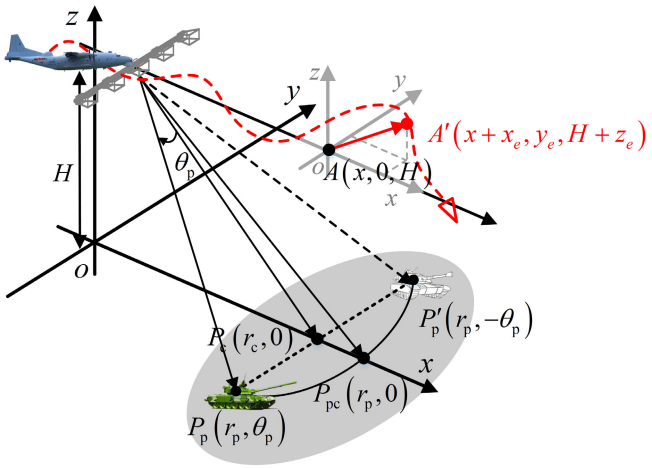


Fig. 1. SAR imaging geometry with high squint angle and wide beam.

From (1) to (5), it becomes evident that FLSAR imaging differs from side-look SAR imaging (SLSAR) in three distinct ways. First, there is the issue of left–right aliasing. Equations (1)–(3) reveal that the slant range from target $P_{\mathrm{p}}(r_{\mathrm{p}}, \theta_{\mathrm{p}})$ is the same as the symmetric target $P'_{\mathrm{p}}(r_{\mathrm{p}}, -\theta_{\mathrm{p}})$ when there is no deviation in the cross-range direction y_e. Second, motion errors have a significant spatial variation, particularly in y_e. The motion error in the Y-direction is opposite in the left and right scenes. In addition, (5) shows that the Doppler modulation rate changes with azimuth. This makes the traditional method of azimuth matching filtering ineffective. As a result, when developing FLSAR imaging algorithms, it is crucial to consider the typical processing procedures in SLSAR imaging and address challenges, such as left–right aliasing, elimination of azimuth-variant Doppler modulation, and motion compensation. These challenges are even more pronounced in high-resolution wide-beam FLSAR.

To effectively resolve the issue of left–right aliasing, an antenna array is employed. The multichannel baseband signal within the range wavenumber domain can be represented

$$\mathbf{s}_e^0(k_{\mathrm{r}}, x, r_{\mathrm{p}}, \theta_{\mathrm{p}}) = \mathbf{a}(\theta_{\mathrm{p}}) \cdot s_e^0(k_{\mathrm{r}}, x, r_{\mathrm{p}}, \theta_{\mathrm{p}}) \tag{6}$$

where $\mathbf{a}(\theta_{\mathrm{p}})$ is the array-oriented vector (AOV) involving θ_{p}, and

$$\mathbf{a}(\theta_{\mathrm{p}}) = \left[1, \exp\left(j2\pi \frac{d \sin\theta_{\mathrm{p}}}{\lambda}\right), \ldots, \exp\left(j2\pi \frac{(M-1)d \sin\theta_{\mathrm{p}}}{\lambda}\right)\right]^{\mathrm{T}}. \tag{7}$$

Based on the array signal model in (6), a lot of left–right Doppler aliasing solution algorithms [26] are developed to reconstruct the unaliasing FLSAR images.

However, as far as the author knows, more literature on high-resolution imaging and spatial-variant MOCO algorithms for wide-beam FLSAR is currently needed. Therefore, this article aims to develop a new PFA that integrates spatial-variant MOCO, aiming to achieve high-resolution wide-beam FLSAR imaging.

III. Forward-Look PFA

This section presents a new algorithm called the forward-looking SAR polar format algorithm (FL-PFA) for high-resolution wide-beam FLSAR imaging. The algorithm uses a time–spatial–time hybrid scheme to address the challenge of spatiotemporal coupling. This method enables the algorithm to generate FLSAR images gradually, starting from coarse-to-fine resolution, without aliasing and with clear focus.

A. Mathematical Model of FL-PFA

After the radio frequency (RF) signal has been demodulated into a baseband signal, as shown in (3), the "azimuth deramp" is implemented in the range wavenumber domain. The reference function $H_e^0(k_{\mathrm{r}}, x, r_{\mathrm{c}}, 0)$ is given by

$$H_e^0(k_{\mathrm{r}}, x, r_{\mathrm{c}}, 0) = \exp\left[jk_{\mathrm{r}} \cdot R_e(x, r_{\mathrm{c}}, 0)\right] \tag{8}$$

where the instantaneous range $R_e(x, r_{\mathrm{c}}, 0)$ from the actual APC to the scene center reference point $P_{\mathrm{c}}(r_{\mathrm{c}}, 0)$ is given by

$$R_e(x, r_{\mathrm{c}}, 0) = \sqrt{(x + x_e - x_{\mathrm{c}})^2 + y_e^2 + (H + z_e)^2} \tag{9}$$

where

$$x_{\mathrm{c}} = \sqrt{(r_{\mathrm{c}})^2 - H^2}. \tag{10}$$

Then, the deramped signal in fast-time domain is given by

$$\begin{aligned} &s_e^1(k_{\mathrm{r}}, x, r_{\mathrm{p}}, \theta_{\mathrm{p}}) \\ &\quad = \exp\left[-jk_{\mathrm{r}} \cdot \left(R_e(x, r_{\mathrm{p}}, \theta_{\mathrm{p}}) - R_e(x, r_{\mathrm{c}}, 0)\right)\right] \\ &\quad = \exp\left[-jk_{\mathrm{r}} \cdot dR_e(x, r_{\mathrm{p}}, \theta_{\mathrm{p}})\right]. \end{aligned} \tag{11}$$

The instantaneous differential range $dR_e(x, \theta_{\mathrm{p}}, r_{\mathrm{p}})$ is given by

$$dR_e(x, r_{\mathrm{p}}, \theta_{\mathrm{p}}) = R_e(x, r_{\mathrm{p}}, \theta_{\mathrm{p}}) - R_e(x, r_{\mathrm{c}}, 0). \tag{12}$$

After performing Taylor series expansion at $\theta_{\mathrm{p}} = 0$ and neglecting the three-order and high-order terms, (12) is rewritten as follows:

$$\begin{aligned} dR_e(x, r_{\mathrm{p}}, \theta_{\mathrm{p}}) \approx\ &\alpha_0(x, r_{\mathrm{p}}, r_{\mathrm{c}}) \\ &+ \alpha_1(x, r_{\mathrm{p}}) \cdot \theta_{\mathrm{p}} + \alpha_2(x, r_{\mathrm{p}}) \cdot \theta_{\mathrm{p}}^2 \end{aligned} \tag{13}$$

where the zero-order term coefficient $\alpha_0(x, r_{\mathrm{p}}, r_{\mathrm{c}})$ is given by the following equation:

$$\alpha_0(x, r_{\mathrm{p}}, r_{\mathrm{c}}) = R_e(x, r_{\mathrm{p}}, 0) - R_e(x, r_{\mathrm{c}}, 0). \tag{14}$$

In addition, the linear term coefficient $\alpha_1(x, r_{\mathrm{p}})$ is given

$$\alpha_1(x, r_{\mathrm{p}}) = -y_e \cdot \frac{r_{\mathrm{p}}}{R_e(x, r_{\mathrm{p}}, 0)} \tag{15}$$

and the two-order term coefficient $\alpha_2(x, r_{\mathrm{p}})$ is

$$\alpha_2(x, r_{\mathrm{p}}) = \frac{r_{\mathrm{p}}^2}{2}\left[\frac{x + x_e}{x_{\mathrm{pc}} \cdot R_e(x, r_{\mathrm{p}}, 0)} - \frac{y_e^2}{R_e^3(x, r_{\mathrm{p}}, 0)}\right] \tag{16}$$

where $x_{\mathrm{pc}} = (r_{\mathrm{p}}^2 - H^2)^{1/2}$.

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B. Coarse-Focused Image Forming

From (12) to (15), we can see that zero-order term $\alpha_0(x, r_\mathrm{p}, r_\mathrm{c})$ contains the bulk of range cell migration and the range-variant motion errors. The linear term coefficient $\alpha_1(x, r_\mathrm{p})$ involves the cross-range trajectory deviation y_e, resulting in the spatial-variant motion error. It is clear that the azimuth phase errors brought by y_e are opposite in the left and right scenes, which means that the motion errors brought by y_e should be compensated after left–right Doppler aliasing solving. The two-order term coefficient $\alpha_2(x, r_\mathrm{p})$ represents the azimuth angle wavenumber which is different from that of SLSAR [27], indicating the low-Doppler modulation rate characteristic of FLSAR than SLSAR.

Then, the "range walk correction" is implemented based on (13). From (13) to (16), one can see that the zero-order term $\alpha_0(x, r_\mathrm{p}, r_\mathrm{c})$ contains the majority of range cell migration. We expand (14) as (17), as shown at the bottom of the next page. The first term is the range difference between $P_\mathrm{p}(r_\mathrm{p}, 0)$ and $P_\mathrm{c}(r_\mathrm{c}, 0)$. The second term is range walk and the third term is range curvature.

In practice, the trajectory deviations are much smaller than the reference range r_c and target range r_p, i.e., $x_e, y_e, z_e \le r_\mathrm{c}$. As a result, the range difference can be further approximated as follows:

$$\begin{aligned}\Delta r_\mathrm{pc} &= \sqrt{\left(x_e - x_\mathrm{pc}\right)^2 + y_e^2 + (H + z_e)^2} \\ &\quad - \sqrt{(x_e - x_\mathrm{c})^2 + y_e^2 + (H + z_e)^2} \\ &\approx r_\mathrm{p} - r_\mathrm{c}. \end{aligned} \tag{18}$$

At the same time, the range walk term is rewritten as follows:

$$\begin{aligned}Rw_\mathrm{pc} &= \frac{x_e - x_\mathrm{pc}}{\sqrt{\left(x_e - x_\mathrm{pc}\right)^2 + y_e^2 + (H + z_e)^2}} \cdot x \\ &\quad - \frac{x_e - x_\mathrm{c}}{\sqrt{(x_e - x_\mathrm{c})^2 + y_e^2 + (H + z_e)^2}} \cdot x \\ &\approx -\left(\varphi_\mathrm{p} - \varphi_\mathrm{c}\right) \cdot x. \end{aligned} \tag{19}$$

where $\varphi_\mathrm{p} = \mathrm{asin}(x_\mathrm{pc}/r_\mathrm{p})$ and $\varphi_\mathrm{c} = \mathrm{asin}(x_\mathrm{c}/r_\mathrm{c})$ are the pitch angles of target point $P_\mathrm{p}(r_\mathrm{p}, 0)$ and reference point $P_\mathrm{c}(r_\mathrm{c}, 0)$, respectively.

In practice, the range curvature term can always be ignored because it is far less than the range resolution. In order to illustrate the approximation rationality, the simulation results with an X-band SAR system are shown in Fig. 2. The approximation error of our algorithm is the residual range curvature shown in Fig. 2(b), and the maximum range curvature is −0.12 m. It is significantly smaller than the range resolution of FLSAR, typically at the submeter level. However, for the FLSAR with a higher range resolution, the range curvature correction will be considered in our future work.

As a result, the range walk can be corrected by the keystone transform (KT) [28], which can be represented as the azimuth position variable substitution

$$k_\mathrm{rc} \cdot x = k_\mathrm{r} \cdot x' \tag{20}$$

where $k_\mathrm{rc} = 4\pi f_\mathrm{c}/c$ is the range wavenumber center. After range walk correction, the signal in (11) becomes

$$\begin{aligned}s_e^2\left(k_\mathrm{r}, x, r_\mathrm{p}, \theta_\mathrm{p}\right) &\approx \exp\left[-jk_\mathrm{r} \cdot \left(r_\mathrm{p} - r_\mathrm{c}\right)\right] \\ &\quad \cdot \exp\left[-jk_\mathrm{rc} \cdot dR_e\left(x, r_\mathrm{p}, \theta_\mathrm{p}\right)\right]. \end{aligned} \tag{21}$$

Herein, the azimuth position variable x' is replaced by x for simplicity. We neglect the impact brought by KT on the linear and quadratic terms because they are far smaller than the zero-order term. The corresponding range-compressed domain signal is

$$s_e^2\left(x, r_\mathrm{p}, \theta_\mathrm{p}\right) = \mathrm{sinc}_r\left(r_\mathrm{p}\right) \exp\left[-jk_\mathrm{rc} dR_e\left(x, r_\mathrm{p}, \theta_\mathrm{p}\right)\right] \tag{22}$$

where $\mathrm{sinc}_r(r_\mathrm{p}) = \mathrm{sinc}(r - r_\mathrm{p})$ is the range envelope.

In addition, from (19), one can see that the Doppler centroid involves the pitch angle of the target, and it is a linear function of φ_p. The Doppler centroid is given by

$$f_\mathrm{dc}\left(\varphi_\mathrm{p}\right) = \frac{2v}{\lambda} \cdot \sin\varphi_\mathrm{p} = \frac{2v}{\lambda} \cdot \frac{\sqrt{r_\mathrm{p}^2 - H^2}}{r_\mathrm{p}}. \tag{23}$$

The linear Doppler centroid migration will cause geometric distortion. Here, we correct the Doppler centroid migration while compensating for the range-variant motion errors. The compensation function can be given by

$$H_e^1\left(x, r_\mathrm{p}, 0\right) = \exp\left[jk_\mathrm{rc} \cdot \alpha_0\left(x, r_\mathrm{p}, r_\mathrm{c}\right)\right]. \tag{24}$$

After range-variant MOCO, the differential range in (12) becomes

$$\begin{aligned}dR_e\left(x, r_\mathrm{p}, \theta_\mathrm{p}\right) &= R_e\left(x, r_\mathrm{p}, \theta_\mathrm{p}\right) - R_e\left(x, r_\mathrm{p}, 0\right) \\ &\approx \alpha_1\left(x, r_\mathrm{p}\right) \cdot \theta_\mathrm{p} + \alpha_2\left(x, r_\mathrm{p}\right) \cdot \theta_\mathrm{p}^2. \end{aligned} \tag{25}$$

We can see that range-dependent motion errors in the zeros-order term have been compensated completely after MOCO by (24). In addition, the Doppler centroid migration is also removed, which means that the image distortion has been corrected.

After range-dependent MOCO, the signal in (22) becomes

$$\begin{aligned}s_e^3\left(x, r_\mathrm{p}, \theta_\mathrm{p}\right) &= \exp\left[-jk_\mathrm{rc} \cdot dR_e\left(x, r_\mathrm{p}, \theta_\mathrm{p}\right)\right] \\ &\approx \exp\left[-jk_\mathrm{rc} \cdot \alpha_1\left(x, r_\mathrm{p}\right) \cdot \theta_\mathrm{p}\right] \\ &\quad \cdot \exp\left[-jk_\mathrm{rc} \cdot \alpha_2\left(x, r_\mathrm{p}\right) \cdot \theta_\mathrm{p}^2\right]. \end{aligned} \tag{26}$$

Herein, the range envelope of the signal is ignored to simplify the analysis.

From (26), the first exponential term is the azimuth-variant phase error, and the second exponential term will be defined as the azimuth angle wavenumber. In order to delete the azimuth-variant phase errors brought by y_e, the left–right Doppler aliasing must be solved before azimuth-variant MOCO. For this reason, the coarse-focused FLSAR image is generated by fast Fourier transform (FFT) along slow time, and it is given by

$$\begin{aligned}I_e\left(r_\mathrm{p}, \theta_\mathrm{p}\right) &= I_e\left(r_\mathrm{p}, f_\mathrm{d,p}\right) = \mathrm{FFT}_x\left[s_e^3\left(r, x, r_\mathrm{p}, \theta_\mathrm{p}\right)\right] \\ &= \mathrm{sinc}\left(r - r_\mathrm{p}\right) \cdot \mathrm{sinc}\left(f_\mathrm{d} - f_\mathrm{d,p}\right) \otimes \Psi_{f_\mathrm{d}}\left(\theta_\mathrm{p}\right). \end{aligned} \tag{27}$$

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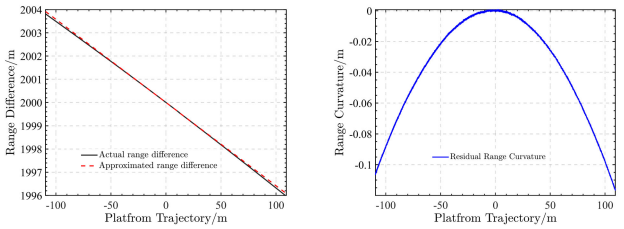


Fig. 2. Approximation error between actual range difference and our expression. (a) Actual range difference versus our linear approximation result. (b) Residual range curvature.

where $\otimes$ denoted the convolutional operation. Moreover $\Psi(\theta_{\mathrm{p}})$ is the azimuth response function of the nonlinear azimuth phase, which contains the phase error and the high-order phase brought by the first and second exponential terms in (26), respectively.

Similarity, the coarse-focused image of the symmetrical point $P_{\mathrm{p}}(r_{\mathrm{p}}, -\theta_{\mathrm{p}})$ is

$$\begin{aligned} I_e\left(r_{\mathrm{p}}, -\theta_{\mathrm{p}}\right) &= I_e\left(r_{\mathrm{p}}, f_{\mathrm{d,p}}\right) \\ &= \mathrm{sinc}\left(r - r_{\mathrm{p}}\right) \cdot \mathrm{sinc}\left(f_{\mathrm{d}} - f_{\mathrm{d,p}}\right) \otimes \Psi_{f_{\mathrm{d}}}\left(-\theta_{\mathrm{p}}\right). \end{aligned} \tag{28}$$

Furthermore, the aliasing coarse-focused FLSAR image is

$$\begin{aligned} I_e^{\mathrm{amb}}\left(r_{\mathrm{p}}, f_{\mathrm{d,p}}\right) &= I_e^{\mathrm{amb}}\left(r_{\mathrm{p}}, \theta_{\mathrm{p}}\right) \\ &= I_e\left(r_{\mathrm{p}}, \theta_{\mathrm{p}}\right) + I_e\left(r_{\mathrm{p}}, -\theta_{\mathrm{p}}\right) \end{aligned} \tag{29}$$

Based on (29), the multichannel signal model of aliasing coarse-focused image can be given by

$$\begin{aligned} \mathbf{s}_e^4\left(r_{\mathrm{p}}, f_{\mathrm{d,p}}\right) &= \mathbf{s}_e^4\left(r_{\mathrm{p}}, \theta_{\mathrm{p}}\right). \\ &= \mathbf{a}\left(\theta_{\mathrm{p}}\right) \cdot I_e\left(r_{\mathrm{p}}, \theta_{\mathrm{p}}\right) + \mathbf{a}\left(-\theta_{\mathrm{p}}\right) \cdot I_e\left(r_{\mathrm{p}}, -\theta_{\mathrm{p}}\right). \end{aligned} \tag{30}$$

The left–right Doppler aliasing solving is implemented on the aliasing coarse-focused image $I_e^{\mathrm{amb}}(r_{\mathrm{p}}, f_{\mathrm{d,p}})$. The left–right aliasing solution based on beam forming is given by [26]

$$\begin{bmatrix} I_e\left(r_{\mathrm{p}}, \theta_{\mathrm{p}}\right) \\ I_e\left(r_{\mathrm{p}}, -\theta_{\mathrm{p}}\right) \end{bmatrix} = \mathbf{w}^{\mathrm{H}}\left(f_{\mathrm{d,p}}\right) \cdot \mathbf{s}_e^4\left(r_{\mathrm{p}}, f_{\mathrm{d,p}}\right). \tag{31}$$

The weight vector $\mathbf{w}(f_{\mathrm{d,p}})$ can be obtained by

$$\begin{aligned} \mathbf{w}\left(f_{\mathrm{d,p}}\right) &= \mathbf{w}\left(\theta_{\mathrm{p}}\right) \\ &= \mathbf{A}\left(\theta_{\mathrm{p}}\right) \cdot \left[\mathbf{A}^{\mathrm{H}}\left(\theta_{\mathrm{p}}\right) \cdot \mathbf{A}\left(\theta_{\mathrm{p}}\right)\right]^{-1} \end{aligned} \tag{32a}$$

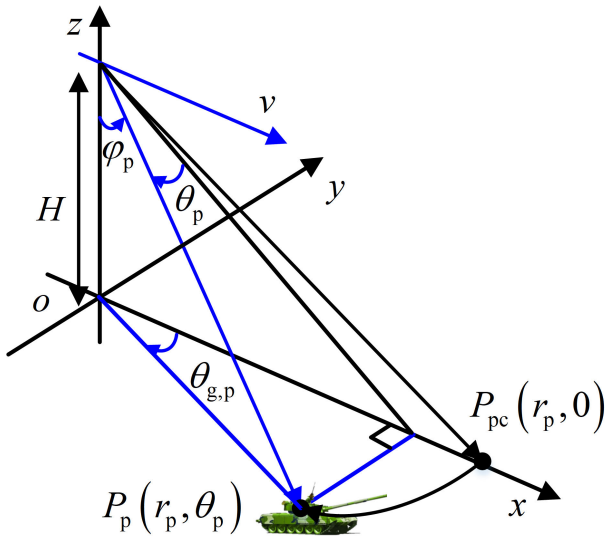


Fig. 3. Geometric relationship of FLSAR spatial–time mapping.

$$\mathbf{A}\left(\theta_{\mathrm{p}}\right) = \left[\mathbf{a}\left(\theta_{\mathrm{p}}\right), \mathbf{a}\left(-\theta_{\mathrm{p}}\right)\right]. \tag{32b}$$

The relationship between $f_{\mathrm{d,p}}$ and θ_{p}, i.e., the FLSAR space–time mapping is required to calculate the weight vector $\mathbf{w}(\theta_{\mathrm{p}})$, because the coarse-focused image is in the Doppler domain. According to the geometric relationship in Fig. 3, The FLSAR space–time mapping is given by

$$\sin\theta_{\mathrm{p}} = \sin\varphi_{\mathrm{p}} \cdot \sin\theta_{g,p} \tag{33a}$$

$$\sin\varphi_{\mathrm{p}} = \frac{H}{r_{\mathrm{p}}} \tag{33b}$$

$$\sin\theta_{g,p} = \sqrt{1 - \left(\frac{f_{\mathrm{d,p}} \cdot \lambda}{2v \cdot \sin\varphi_{\mathrm{p}}}\right)^2}. \tag{33c}$$

In addition, (31) is precise without channel amplitude and phase error; otherwise, the channel correction, adaptive beamforming, and compressed sensing methods can be applied to obtain more robust results.

After left–right Doppler aliasing solving, the nonaliasing signal is given by

$$\begin{aligned} s_e^5\left(x, r_{\mathrm{p}}, \theta_{\mathrm{p}}\right) &= \exp\left(-j4\pi\, f_{\mathrm{c}} \frac{\alpha_1\left(x, r_{\mathrm{p}}\right)}{c} \cdot \theta_{\mathrm{p}}\right) \\ &\quad \cdot \exp\left(-j4\pi\, f_{\mathrm{c}} \frac{\alpha_2\left(x, r_{\mathrm{p}}\right)}{c} \cdot \theta_{\mathrm{p}}^2\right). \end{aligned} \tag{34}$$

$$\begin{aligned} \alpha_0\left(x, r_{\mathrm{p}}, r_{\mathrm{c}}\right) \approx & \left[\sqrt{(x_e - x_{\mathrm{pc}})^2 + y_e^2 + (H + z_e)^2} - \sqrt{(x_e - x_{\mathrm{c}})^2 + y_e^2 + (H + z_e)^2}\right] \\ & + \left[\frac{x_e - x_{\mathrm{pc}}}{\sqrt{(x_e - x_{\mathrm{pc}})^2 + y_e^2 + (H + z_e)^2}} - \frac{x_e - x_{\mathrm{c}}}{\sqrt{(x_e - x_{\mathrm{c}})^2 + y_e^2 + (H + z_e)^2}}\right] \cdot x \\ & + \frac{1}{2}\left[\frac{\sqrt{(x_e - x_{\mathrm{pc}})^2 + y_e^2 + (H + z_e)^2} - (x_e - x_{\mathrm{pc}})^2}{\sqrt[3/2]{(x_e - x_{\mathrm{pc}})^2 + y_e^2 + (H + z_e)^2}} - \frac{\sqrt{(x_e - x_{\mathrm{c}})^2 + y_e^2 + (H + z_e)^2} - (x_e - x_{\mathrm{c}})^2}{\sqrt[3/2]{(x_e - x_{\mathrm{c}})^2 + y_e^2 + (H + z_e)^2}}\right] \cdot x^2. \end{aligned} \tag{17}$$

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C. Fine-Focused Image Forming

1) 2-D Spatial-variant MOCO: Regarding MOCO, there are two crucial differences between multichannel forward-looking SAR and conventional side-looking SAR. The first distinction is the presence of left–right Doppler aliasing, while the second is the occurrence of opposite spatial-variant phase errors for left and right scenes. Unlike side-looking SAR, these two characteristics make it impossible to simultaneously compensate for motion errors in both left and right scenes, significantly hindering the precise azimuth-variant MOCO in multiple-channel forward-looking SAR imaging. Therefore, this article implements spatial-variant MOCO after solving the left–right Doppler aliasing.

From (13) and (15), the azimuth phase error for target $P_{\mathrm{p}}(r_{\mathrm{p}}, \theta_{\mathrm{p}})$ with azimuth angle θ_{p} can be given by

$$\begin{aligned}\phi_e\left(x, r_{\mathrm{p}}, \theta_{\mathrm{p}}\right) &= k_{\mathrm{rc}} \cdot \alpha_1\left(x, r_{\mathrm{p}}\right) \cdot \theta_{\mathrm{p}} \\ &= -k_{\mathrm{rc}} \cdot y_e \cdot \frac{r_{\mathrm{p}}}{R_e\left(x, r_{\mathrm{p}}, 0\right)} \cdot \theta_{\mathrm{p}} \\ &\approx -k_{\mathrm{rc}} \cdot y_e \cdot \frac{r_{\mathrm{p}}}{R_e(x, r_{\mathrm{c}}, 0)} \cdot \theta_{\mathrm{p}}. \end{aligned} \tag{35}$$

In (35), we replace r_{p} in the denominator with r_{c}, while leaving r_{p} in the numerator unchanged. The approximation is appropriate when $\Delta r_{\mathrm{pc}}/r_{\mathrm{c}} \ll 1$. Under this assumption, the phase $\phi_e(x, r_{\mathrm{p}}, \theta_{\mathrm{p}})$ still remains spatially variant in 2-D, but it has now become a linear function of the range variant r_{p} and azimuth angle variant θ_{p}.

Furthermore, $\phi_e(x, r_{\mathrm{p}}, \theta_{\mathrm{p}})$ is also azimuth variant. The classic algorithm for correcting correct these motion errors is the subaperture topography- and aperture-dependent (SATA) algorithm. It addresses spatial-variant phase errors by dividing the full aperture into subapertures and considering the space–time relationship. However, the original SATA algorithm approximates the motion error within each subaperture as a constant, resulting in low accuracy for MOCO. The scale Fourier transform (SFT)-SATA algorithm was developed as an enhanced version of SATA to improve compensation accuracy. It replaces the constant approximation with a linear approximation. However, its direct use in forward-looking SAR imaging is limited because it requires the velocity error to be spatial-independent. Moreover, both the SATA and SFT-SATA algorithms fail to account for the time-varying nature of trajectory deviation. Consequently, they use a fixed length for the subapertures, which may only be optimal in some cases. This article introduces the adaptive SATA algorithm, known as the ASATA, which reduces computational burden while ensuring compensation accuracy.

Now, let us examine the MOCO procedure in a specific subaperture. For the mth subaperture, we assume that the motion error y_e^m can be approximated as a constant velocity error $y_e^m = v_y^m t$. As described in [2], when motion error is present, the Doppler-angle relationship is given

$$f_{d,\mathrm{p}}^e = \frac{2\left(v \sin\varphi_{\mathrm{p}} \cos\theta_{g,\mathrm{p}} + v_y^m \sin\varphi_{\mathrm{p}} \sin\theta_{g,\mathrm{p}}\right)}{\lambda}. \tag{36}$$

One can see from (36) that the Y-axis velocity error v_y^m will bring an additional Doppler error, leading to an alteration in the spatial–time relationship. Moreover, since the spatial–temporal relationship introduced by motion error differs from the ideal scenario, it cannot be corrected directly using SFT-SATA. Fortunately, the precise azimuth angle can be calculated using (36), which enables an interpolation operation to correct this azimuth-variant motion error. Combining (33) and (36), we can express the interpolation process as follows:

$$f_d^e \overset{\text{interp}}{\rightarrow} f_d \Rightarrow \left(v \cos\theta_g + v_y^m \sin\theta_g\right) \sin\varphi \overset{\text{interp}}{\rightarrow} v \sin\varphi \cos\theta_g. \tag{37}$$

After interpolation in (37) the velocity error v_y is eliminated. The constant phase error in the subaperture center is given

$$\begin{aligned}\phi_e^m\left(x^m, r_{\mathrm{p}}, \theta_{\mathrm{p}}\right) = &-k_{\mathrm{rc}} \cdot \hat{y}_e^m \cdot \frac{r_{\mathrm{p}}}{R_e(x, r_{\mathrm{c}}, 0)} \\ &\cdot \arcsin\left[\frac{H}{r_{\mathrm{p}}}\sqrt{1 - \left(\frac{f_{\mathrm{d,p}} \cdot \lambda \cdot r_{\mathrm{p}}}{2v \cdot H}\right)^2}\right] \end{aligned} \tag{38}$$

where $\hat{y}_e^m$ is the Y-axis motion error in the mth subaperture center. Based on (38), the azimuth compensation function for target $P_{\mathrm{p}}(r_{\mathrm{p}}, \theta_{\mathrm{p}})$ can be expressed as follows:

$$H_e^2(t, f_{\mathrm{d}}) = \exp\left\{\begin{array}{l} jk_{\mathrm{rc}} \cdot \hat{y}_e^m \cdot \dfrac{r_{\mathrm{p}}}{R_e\left(x, r_{\mathrm{p}}, 0\right)} \cdot \arcsin \\ \left[\sin\varphi_{\mathrm{p}} \cdot \left(1 - \left(\dfrac{f_{\mathrm{d}} \cdot \lambda}{2v \cdot \sin\varphi_{\mathrm{p}}}\right)^2\right)\right] \end{array}\right\}. \tag{39}$$

After 2-D spatial-variant MOCO, the signal becomes

$$s_e^6\left(x, r_{\mathrm{p}}, \theta_{\mathrm{p}}\right) = \exp\left(-jk_{\mathrm{rc}} \cdot \alpha_2\left(x, r_{\mathrm{p}}\right) \cdot \theta_{\mathrm{p}}^2\right). \tag{40}$$

The ASATA is shown in Fig. 4. The compensation accuracy is directly related to subaperture length. The longer the subaperture length, the higher the angular resolution for Doppler measurements. However, this also increases the approximation error in motion error. On the other hand, a shorter subaperture reduces the approximation error but results in a lower Doppler angular resolution. Therefore, it is essential to carefully optimize the subaperture length to minimize compensation error. According to Fig. 4, the compensation error is given

$$\begin{aligned}\Delta\phi_e^m &= -k_{\mathrm{rc}} \cdot \Delta y_e^m(l) \cdot \frac{r_{\mathrm{c}}}{R_e(x, r_{\mathrm{c}}, 0)} \cdot \frac{\Delta\theta^m}{2} \\ &\approx -k_{\mathrm{rc}} \cdot \Delta y_e^m(l) \cdot \frac{\Delta\theta^m}{2} \end{aligned} \tag{41}$$

where $\Delta y_e^m(l)$ is approximation error of y_e^m, which usually obtained at the edge of the subaperture; l^m is the mth subaperture length. Moreover, based on (33), $\Delta\theta^m$ is

$$\begin{aligned}\Delta\theta^m &= \left\{\sin^{-1}\left[\sqrt{\sin^2\varphi_c - \left(\frac{f_{\mathrm{d}} \cdot \lambda}{2v}\right)^2}\right]\right\}' \Delta f_{\mathrm{d}} \\ &= \frac{-f_{\mathrm{d}}\lambda^2 \csc^2\varphi_c \cdot \Delta f_{\mathrm{d}}}{4v^2\sqrt{1 - \frac{f_{\mathrm{d}}^2\lambda^2\csc^2\varphi_c}{4v^2}}\sqrt{1 - \left(1 - \frac{f_{\mathrm{d}}^2\lambda^2\csc^2\varphi_c}{4v^2}\right)}}. \end{aligned} \tag{42}$$

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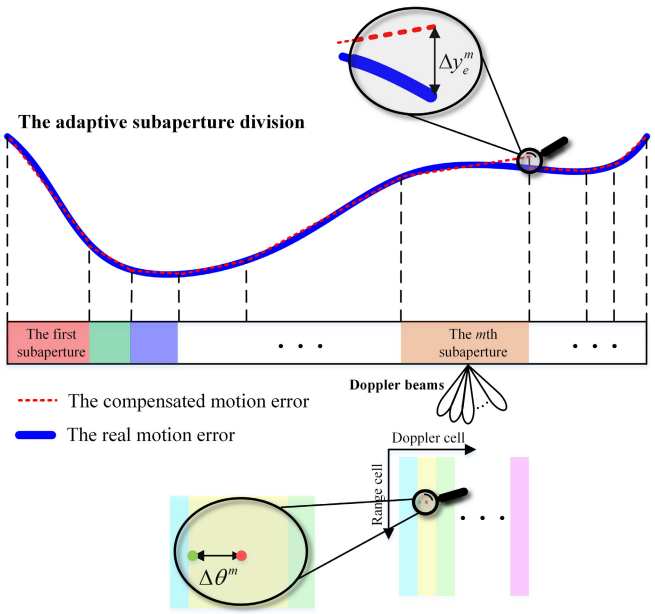


Fig. 4. Schematic of ASATA algorithm.

Then, we have

$$\Delta\theta^m \le \frac{P_{rf}\lambda^2\csc^2\varphi_c \cdot \frac{P_{rf}}{l^m}}{8v^2\sqrt{1-\frac{P_{rf}^2\lambda^2\csc^2\varphi_c}{16v^2}}\sqrt{1-\left(1-\frac{P_{rf}^2\lambda^2\csc^2\varphi_c}{16v^2}\right)}} \tag{43}$$

where P_{rf} is the pulse repeat frequency (PRF). Substituting (43) into (41), we have

$$\Delta\phi_e^m = \frac{k_{\rm rc}\cdot\Delta y_e^m(l)\cdot P_{rf}\lambda^2\csc^2\varphi_c\cdot\frac{P_{rf}}{l}}{8v^2\sqrt{1-\frac{P_{rf}^2\lambda^2\csc^2\varphi_c}{16v^2}}\sqrt{1-\left(1-\frac{P_{rf}^2\lambda^2\csc^2\varphi_c}{16v^2}\right)}}. \tag{44}$$

The optimal subaperture length l can be obtained by optimizing the $\Delta\phi_e^m$ minimization, i.e.,

$$\begin{aligned}&\min_{l^m}\ \Delta\phi_e^m\\ &\text{s.t. } l^m \ge l_{\min}^m\end{aligned} \tag{45}$$

where $l_{\min}^m$ is the smallest subaperture length. In this article, we use the forward–backward search algorithm [29] to solve the optimization problem mentioned above due to its simplicity and efficiency.

2) Azimuth-Variant Angle-Wavenumber Resampling: We can see from (40) that the azimuth signal can be regarded as a single frequency signal in the azimuth angle wavenumber domain, with

$$k_{\theta,e} = k_{\rm rc}\cdot\alpha_2(x, r_{\rm p}). \tag{46}$$

Herein, $k_{\theta,e}$ is defined as quadratic angle wavenumber containing motion errors because it's coefficient is $\theta_{\rm p}^2$ rather than $\theta_{\rm p}$. Then, substituting (46) into (40), we can obtain the following equation:

$$s_e^6(k_{\theta,e},\theta_{\rm p}) = \text{sinc}_r(r_{\rm p})\cdot\exp\left(-jk_{\theta,e}\cdot\theta_{\rm p}^2\right). \tag{47}$$

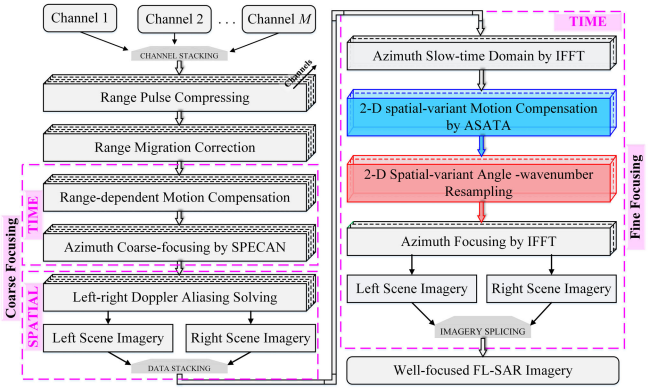


Fig. 5. Flowchart of the proposed FL-PFA algorithm.

From (47), a well-focused, forward-looking SAR imagery can be generated by performing an inverse FFT (IFFT) operation in the quadratic angle wavenumber domain. However, the well-focused imagery is given by

$$\begin{aligned}I(r_{\rm p},\theta_{\rm p}) &= \text{IFFT}_{k_\theta}\left[s_e^6(k_\theta,\theta_{\rm p})\right]\\ &= \text{sinc}_r(r_{\rm p})\cdot\text{sinc}_\theta\left(\theta_{\rm p}^2\right)\end{aligned} \tag{48}$$

where $\text{sinc}_\theta(\theta_{\rm p}^2) = \text{sinc}(\theta-\theta_{\rm p}^2)$ is the azimuth envelope.

In summary, the entire flowchart of FL-PFA is shown in Fig. 5.

IV. Simulation And Real Data Experiments

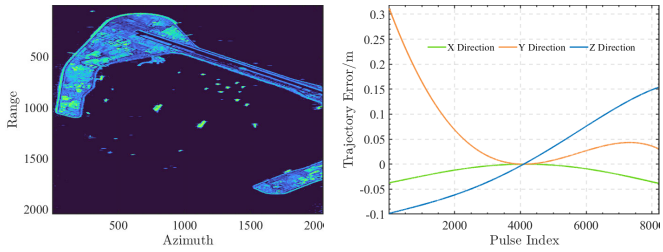
In this section, we will present the results of both simulated and real-measured FLSAR experiments using the developed algorithm. We will thoroughly examine the effectiveness and superiority of FL-PFA through extensive comparison experiments. In addition, image contrast [30] will be used to quantitatively assess the quality of FLSAR images in both simulation and real-measured experiments.

A. Experiments With Simulated Data

In this section, we use the proposed FL-PFA to process simulated multichannel FLSAR echo data of a surface target and verify its effectiveness. The scene under investigation is an isolated island with a range width of 737 m and an azimuth width of approximately 3600 m, as shown in Fig. 6(a). In addition, Fig. 6(b) depicts the 3-D trajectory deviations utilized in the simulation experiments. It is worth noting that the Y-axis motion error is significantly pronounced, with a maximum deviation of 0.32 m, which indicates that spatial-variant motion error is inevitable. Moreover, the critical parameters of the FLSAR simulation system are listed in Table I. The transmission bandwidth is 410 MHz, which enables a range resolution of approximately 0.32 m. In addition, the system achieves an azimuth resolution of about 1 m with an azimuth beamwidth of approximately 13.3°, which indicates its high resolution and wide beam capabilities.

First, the echo data without any motion errors is generated to examine the focusing performance of FL-PFA. We compare FL-PFA with the SPECAN algorithm in [2] and the back-projection algorithm (BPA) [25]. The results are presented

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(a) (b)

Fig. 6. Elements of the simulation experiment. (a) Original imaging scene. (b) Trajectory deviations in the simulated experiment.

TABLE I
SIMULATION SAR SYSTEM PARAMETERS

Parameter	Value	Parameter	Value
Carrier frequency	35 GHz	Pulse repeat frequency	1.5 kHz
Band width	410 MHz	Azimuth beam width	13.3 deg
Platform height	2.15 km	Channel Number	9
The center range	15.5 km	Platform velocity	69.26 m/s
The range samples	4096	The number of pulses	8192

in Fig. 7(a)–(c). It is discernible that the imagery procured by the SPECAN exhibits a comparatively elevated level of speckle-noise compared with other algorithms. Furthermore, a noticeable blurriness is observed, specifically around infrastructural contours such as roads and buildings. Conversely, both BPA and FL-PFA yield images characterized by sharp focusing, which is evident both at the central and peripheries of the scene. The image contrast values of these results are calculated to quantitatively evaluate the focusing performance, yielding respective figures of 4.6132, 4.6467, and 4.7320. These contrast values demonstrate that FL-PFA possesses superior focusing capabilities, even outperforming BPA.

To further demonstrate the capacity of the proposed algorithm to handle the spatial-variant Doppler modulation rate, three scenes were chosen: scene B in the center and scenes A and C on either side of the image. These scenes are enlarged and shown in Fig. 8(a)–(c). The SPECAN algorithm produces significantly defocused results for the edge scenes A and C. In particular, the road in scene A exhibits a high level of speckle noise due to the defocusing of the island contour lines. Similarly, the building contours in scenes A and C appear blurred for the same reason. In contrast, the central scene B is well-focused, with the airport runway displaying low speckle noise and discernible contours. It can be seen that the SPECAN algorithm is unable to handle the spatial-variant Doppler modulation rate induced by the azimuth wide-beam, resulting in a well-focused center and defocused edges phenomenon. In contrast, the BPA algorithm produces sharp and focused images across all scenes, thanks to its accurate point-to-point convolution. Consequently, the results obtained with the BPA algorithm are considered the baseline in this article. Moreover, it can be observed from Fig. 8(c) that our proposed FL-PFA yields almost identical results to BPA. In scene A, the roads and buildings have distinct outlines. In scene B, the speckle noise level on the runway is low, and the outlines of the dock are visible. In addition, the structures in scene C can

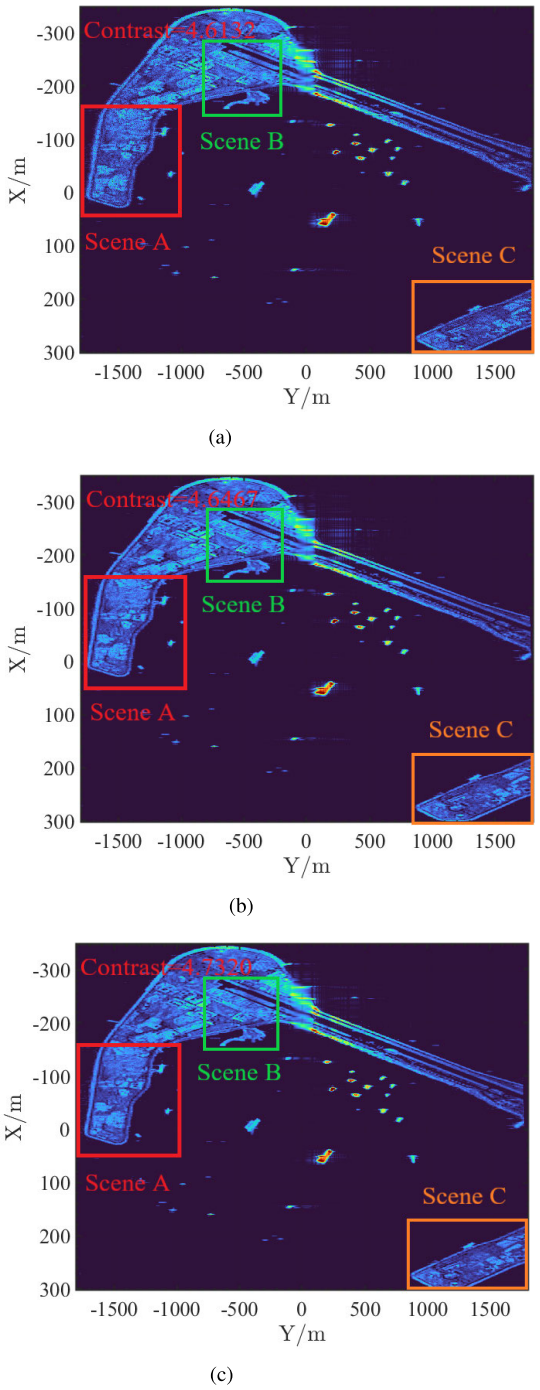


Fig. 7. Comparison of focusing performance of different algorithms (without motion errors). (a) Results from SPECAN. (b) Results from BPA. (c) Results from FL-PFA.

also be discerned. These examples indicate that the proposed algorithm successfully focuses wide-beam FLSAR images. Furthermore, the image contrast values are shown in the upper left corner, providing solid evidence that confirms the suggested algorithm's superiority over SPECAN and demonstrates its similar focusing capabilities to BPA. In conclusion, these results effectively demonstrate the efficiency of FL-PFA in handling high-resolution FLSAR data. It eliminates the remaining azimuth-variant quadratic phase and generates clear and sharp images.

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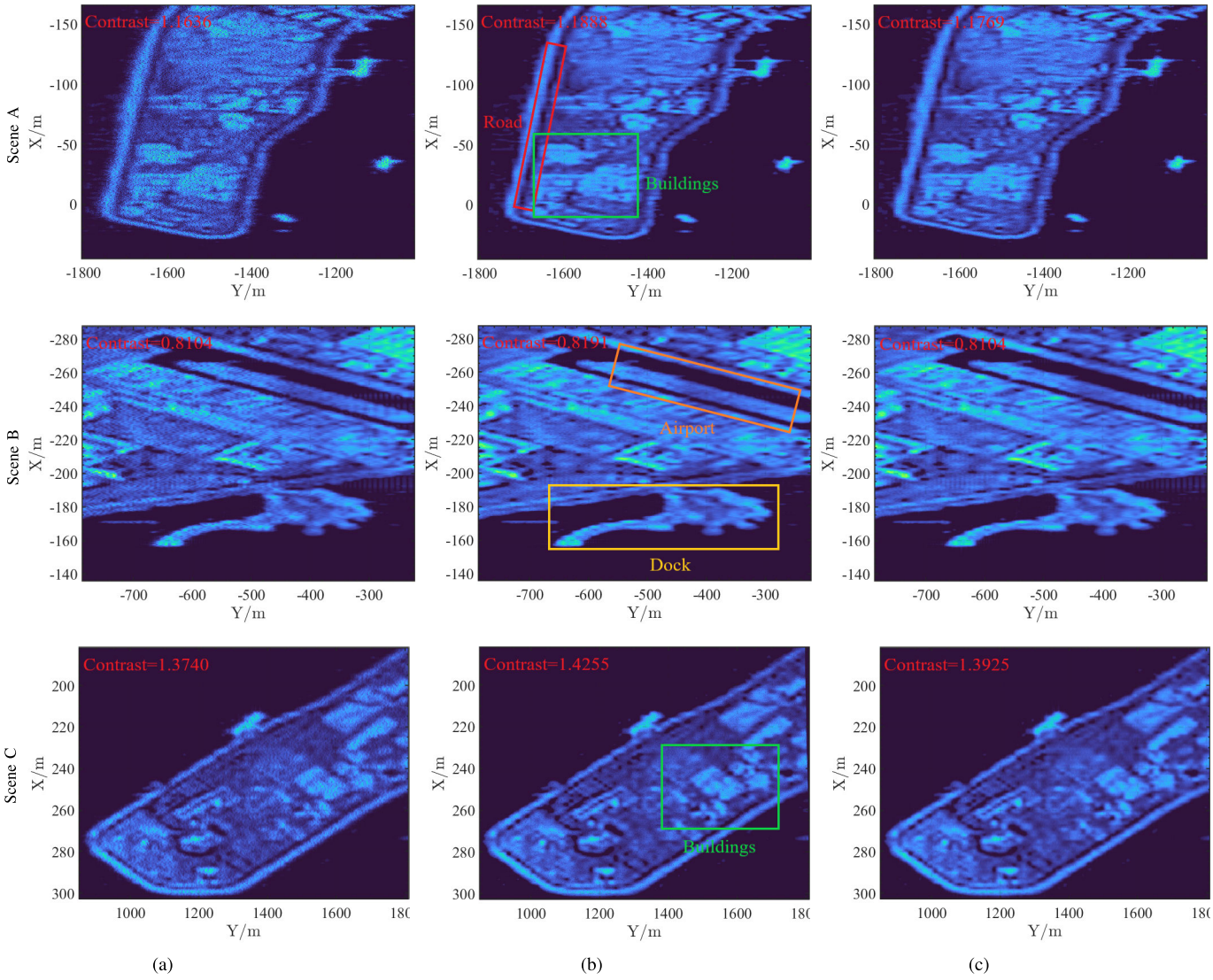


Fig. 8. Subimages of scenes A–C in Fig. 7. (a) Results from SPECAN. (b) Results from BPA. (c) Results from FL-PFA.

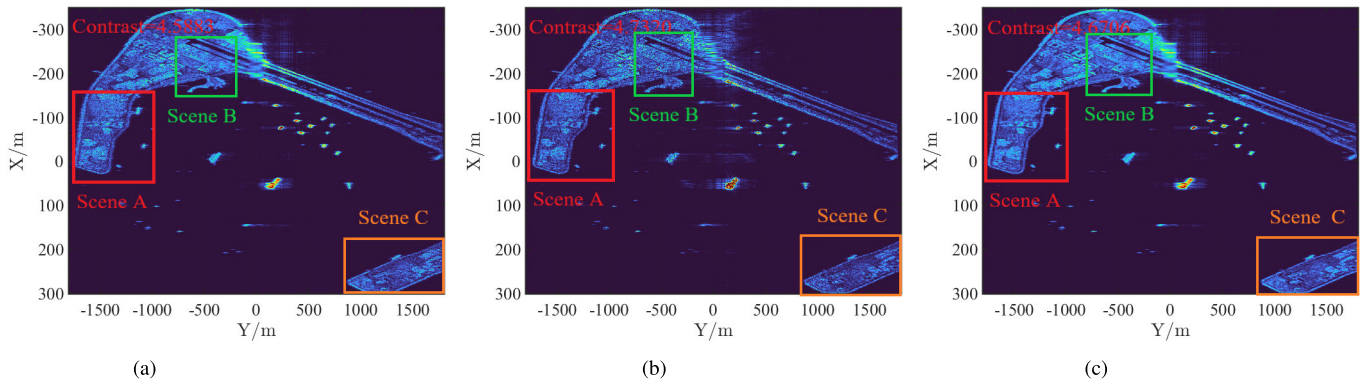


Fig. 9. Comparison of focusing performance of different algorithms (with motion errors). (a) Results from SPECAN + SATA. (b) Results from BPA. (c) Results from FL-PFA.

Second, the FLSAR echo data with motion errors is generated to assess the MOCO performance of FL-PFA thoroughly. The results are illustrated in Fig. 9(a)–(c), respectively. In SPECAN, the original SATA algorithm is employed to rectify the azimuth-variant motion errors. As depicted in Fig. 9(a), the low-precision motion compensation of the SATA algorithm does not effectively rectify spatial-variant motion errors. This exacerbates the image's defocusing effect, ultimately increasing the presence of speckle noise. Conversely, in Fig. 9(b) and (c), the focusing outcomes of BPA, as well

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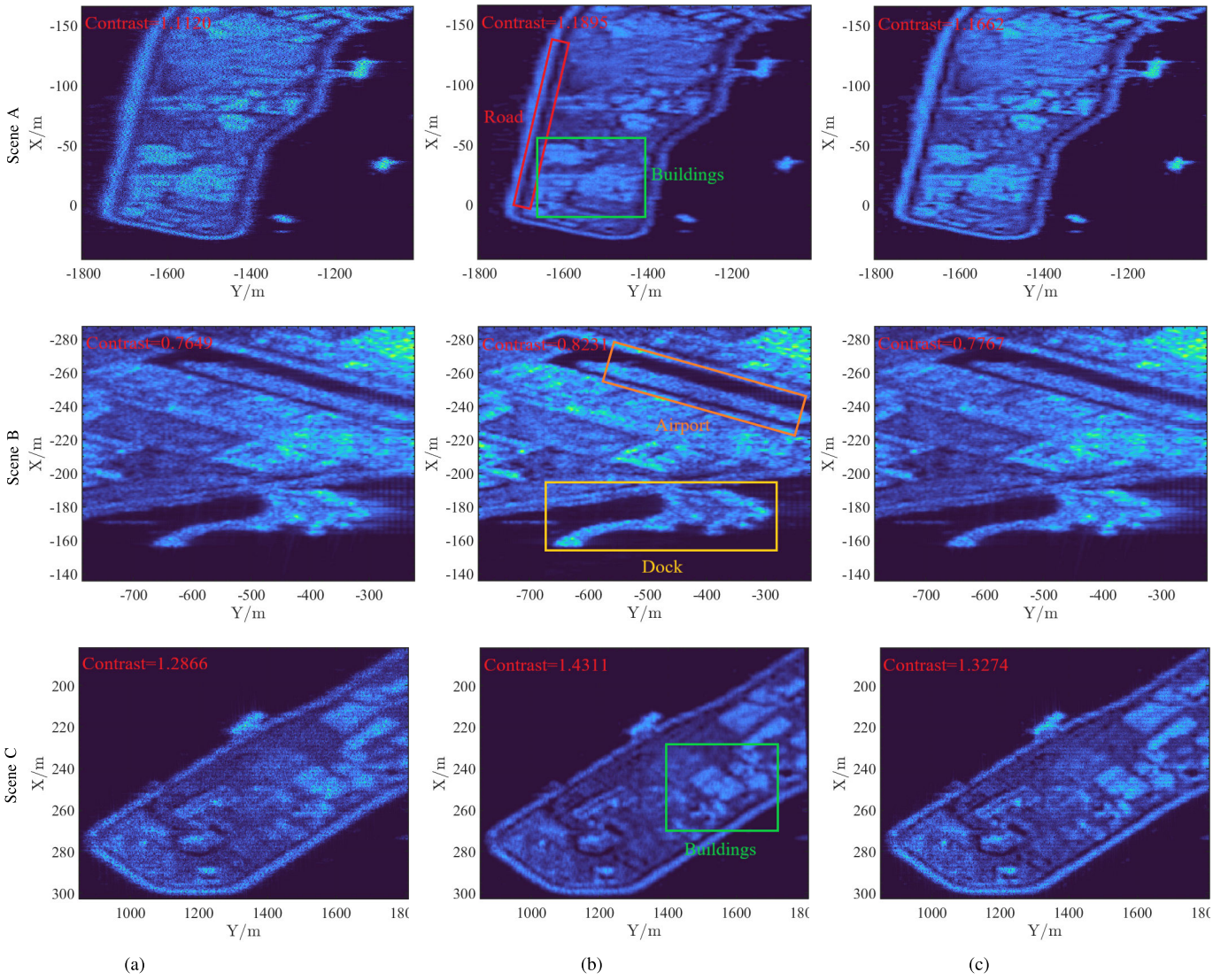


Fig. 10. Subimages of scenes A–C in Fig. 9. (a) Results from SPECAN + SATA. (b) Results from BPA. (c) Results from FL-PFA.

as the proposed algorithm, surpass those of SPECAN. Image contrast is used to quantitatively assess the focusing performance of these algorithms. Upon analyzing the contrast values displayed in the upper left corner of the images, it is clear that the BPA algorithm displays the highest focusing performance, followed by the proposed algorithm. Conversely, the SPECAN algorithm shows the poorest focusing performance.

We still selected the same three scenarios to further evaluate these results and displayed the enlarged subimages in Fig 10(a)–(c). For SPECAN, the defocus issue remains noticeable, especially at the edge scenes, mainly due to the unresolved residual azimuth-variant quadratic phase and the relatively imprecise low-order MOCO. On the contrary, the results of BPA are still satisfactory, as the texture of the buildings and the nearby isolated reefs remain discernible and well-defined. Furthermore, the results from FL-PFA can be observed from Fig. 10(c). By employing accurate spatial-variant MOCO and quadratic phase removal, FL-PFA successfully generates a well-focused FLSAR image with clear island edges and distinct internal textures. In addition, when compared with SPECAN and BPA, the proposed

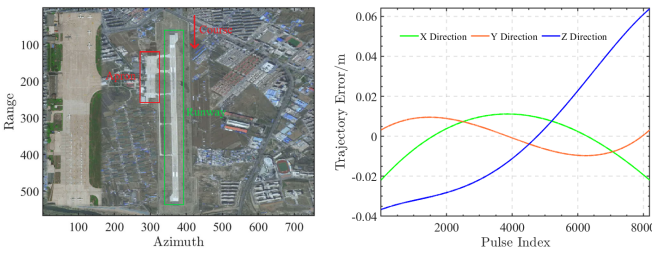


Fig. 11. Elements of the real-measured experiment. (a) Imaging scene. (b) Trajectory deviations in the real-measured experiment.

algorithm's results show a clear decrease in speckle noise. This improvement can be attributed to its better focusing performance and more accurate high-order MOCO. Finally, the image contrast is still displayed in the upper left corner. Based on these values, it is evident that the proposed algorithm performs better than SPECAN and is slightly less effective than BPA.

It is important to point out that both the SPECAN results and the proposed algorithm show that the outlines of the dock in

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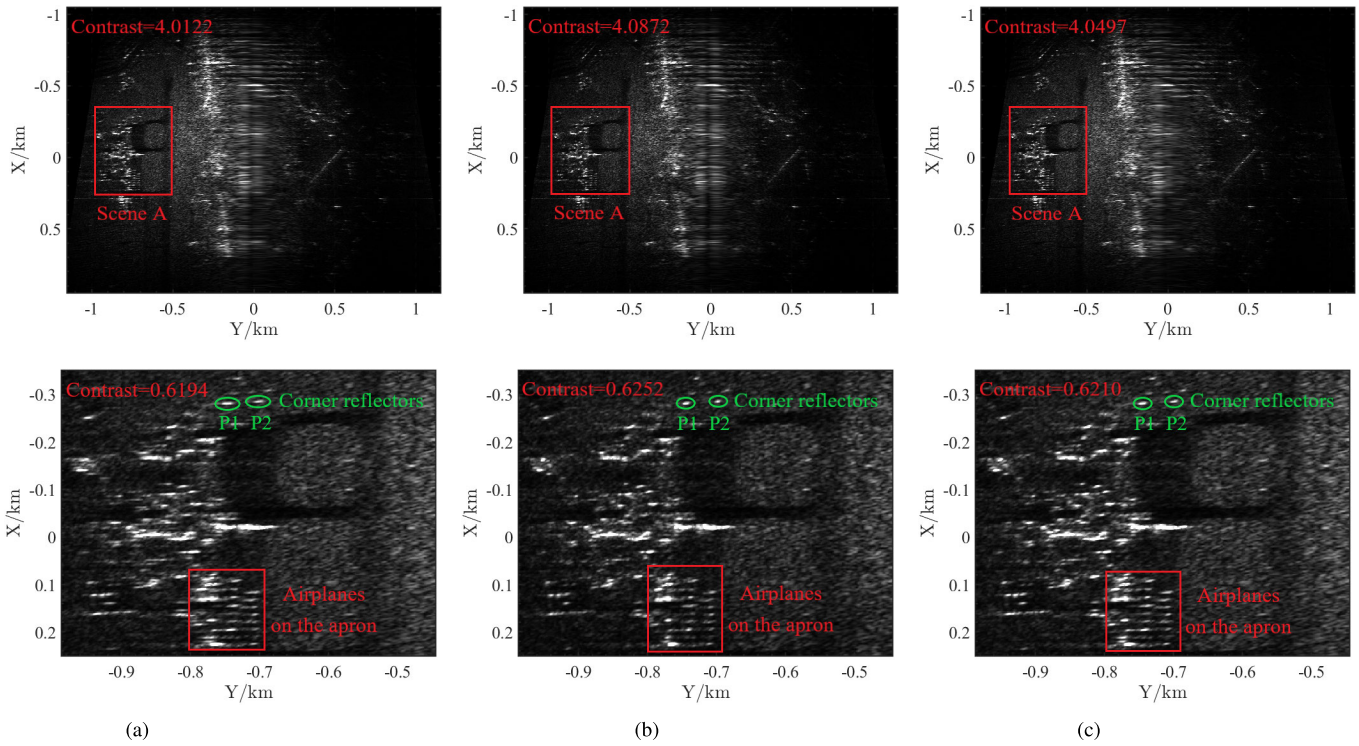


Fig. 12. Comparison of different algorithms. (a) Results from SPECAN + SATA. (b) Results from BPA. (c) Results from FL-PFA.

scene B appear noticeably blurred. This blurring may be due to the decrease in performance of compensating for motion that varies across the scene, which could be caused by the low resolution of the central part of the scene. While the proposed algorithm can ensure accurate compensation by adjusting the aperture length adaptively, it needs to be able to achieve accuracy when dealing with very low resolutions. This limitation should be addressed as an area for future improvement.

In summary, the simulation results thoroughly verify the superior focusing performance and MOCO ability of the proposed FL-PFA for wide-beam FLSAR imaging.

B. Experiments With Real Measured SAR Data

In this section, we will present the experimental results of FLSAR imaging using the FL-PFA algorithm based on actual measurements. We measured the three-axis motion errors using a high-accuracy inertial navigation system (INS) mounted on the platform. The INS provides motion parameters crucial in determining the trajectory deviation during the MOCO process. To mitigate any potential unknown measurement errors in the echo data, we applied an autofocus algorithm [31] before implementing the imaging algorithms.

First, the critical parameters of the actual FLSAR system are listed in Table II. It is important to note that the signal transmission bandwidth is 80 MHz, allowing for a range resolution of approximately 1.6 m. The system also consists of five receiving channels with an array length of 0.012 m.

In addition, Fig. 11(a) shows the optical satellite image of the imaging scene, which includes a general aviation airport. In this image, airplanes can be seen parked on the apron, with several corner reflectors placed around the area.

TABLE II
REAL SAR SYSTEM PARAMETERS

Parameter	Value	Parameter	Value
Carrier frequency	35 GHz	Pulse repeat frequency	5 kHz
Band width	80 MHz	Azimuth beam width	5.0 deg
Platform height	2.81 km	Channel Number	5
The center range	18.0 km	Platform velocity	89.64 m/s
The range samples	2048	The number of pulses	8192

Furthermore, the course is aligned parallel to the runway, and the deviations in trajectory are depicted in Fig. 11(b). Among these deviations, the most significant is the Z-axis motion error, with a maximum deviation of approximately 0.62 m. Meanwhile, the maximum motion errors for the X- and Y-axes are 0.022 and 0.01 m, respectively. It is essential to recognize the impact of these motion errors and take necessary measures to compensate for them accurately.

The imaging results obtained from the proposed FL-PFA algorithm and the other comparison algorithms are presented in Fig. 12(a)–(f). First, it is clear from Fig. 12(a)–(c) that the airports in these results can be easily distinguished. In addition, the airplanes and corner reflectors can also be identified. However, there are significant differences in the focusing performance of the three results, especially for the airplanes and corner reflectors. Intuitively, the result produced by BPA seems to be the most outstanding, followed by FL-PFA, whereas the result obtained from SPECAN clearly shows defocusing. We quantitatively assess the focusing performance by measuring the image contrast to validate this intuition. A higher value indicates better image focus [29]. The contrast values, 4.0122, 4.0872, and 4.0497, demonstrate this point.

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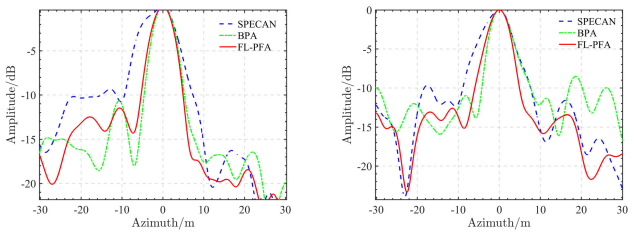


Fig. 13. Azimuth IRFs formed by different algorithms. (a) Corner reflector P1. (b) Corner reflector P2.

TABLE III
IRW, PSLR, AND ISLR VALUES OF POINT TARGETS

Algorithm	Criteria	IRW	PSLR	ISLR
SPECAN	P1	9.3305	-9.3302	-5.4789
	P2	7.2896	-9.6431	-5.1592
BPA	P1	5.5476	-10.6805	-4.0596
	P2	5.3332	-10.9812	-1.7543
FL-PFA	P1	5.7287	-11.4727	-5.6433
	P2	5.4967	-12.5934	-4.9380

Furthermore, to present the imaging results clearly, we have chosen scene A as the comparative reference from the entire image. These scenes are displayed in Fig. 12(d)–(f) after being enlarged. The most noticeable difference is the focusing effect of the corner reflectors. Our proposed FL-PFA produces slightly inferior results than BPA but significantly outperforms SPECAN. FL-PFA exhibits relatively narrow azimuth main lobes. In addition, there are differences in the appearance of the airplanes. With SPECAN, the main lobes of the airplanes are broad, causing the two rows of airplanes to almost merge in the azimuth direction. On the other hand, similar to the results obtained with BPA, our FL-PFA narrows down the width of the main lobes, allowing for a clear distinction between them. Finally, there is a slight disparity in image contrast. The results obtained from BPA and FL-PFA indicate lower levels of imaging noise in the aprons, leading to lower pixel values than those obtained from SPECAN. Consequently, the image contrast of BPA and FL-PFA is higher than that of SPECAN. The contrast values for each sub-image are calculated to support these viewpoints. Specifically, the contrast values are 0.6194, 0.6252, and 0.6210.

In addition, the azimuth impulse response function (IRF) for two corner reflectors is illustrated in Fig. 13(a) and (b). It is evident from the data presented in Fig. 13 that BPA outperforms others in terms of achieving the most narrow main-lobe width along with a low sidelobe level. In comparison, our proposed algorithm demonstrates performance that rivals that of the BPA. Although the main lobe width of FL-PFA is marginally wider than that of BPA, it is noteworthy that it presents a significantly lower sidelobe level. Conversely, SPECAN is observed to yield considerably broader main lobes and a markedly enhanced sidelobe level due to the residual quadratic phase.

The standard measures such as impulse response width (IRW), peak sidelobe ratio (PSLR), and integrated sidelobe ratio (ISLR) serve as the evaluation criteria for quantitatively assessing the focusing capabilities of these algorithms. The outcomes are documented in Table III. The numerical data indicate that the BPA and FL-PFA algorithms substantially outperform the SPECAN in terms of focusing efficiency. Although our proposed algorithm demonstrates a marginally broader main lobe width compared to BPA, it significantly excels in sidelobe performance.

In summary, the simulation and real-measured experiments demonstrate that the proposed FL-PFA exhibits exceptional focusing performance and accurate spatial-variant MOCO capability for high-resolution and wide-beam FLSAR data.

V. CONCLUSION

In this article, we present a PFA designed for high-resolution wide-beam FLSAR imaging, referred to as FL-PFA. Our approach introduces a time–spatial–time processing framework to tackle the challenge of spatial-time coupling and achieve a focused FLSAR image by incorporating MOCO. Within this framework, the FLSAR image is progressively focused from coarse-to-fine resolution. Initially, the coarse-focused image is formed using SPECAN, followed by the application of spatial filtering to eliminate Doppler aliasing. Subsequently, a polar format imaging algorithm is developed to attain a high-resolution wide-beam focused FLSAR image. This algorithm integrates spatial-variant quadratic phase matching and MOCO through a novel angle-wavenumber definition. In addition, the ASATA is combined with FL-PFA to mitigate the residual spatial-variant motion error, effectively eliminating left–right anisotropy. ASATA achieves a balance between compensation accuracy and computational efficiency by adaptively selecting the length of subapertures. Comprehensive experiments are conducted to validate the effectiveness of our approach using both simulated and real-measured FLSAR data.

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REFERENCES

[1] Z. Li, J. Wu, Y. Huang, Z. Sun, and J. Yang, "Ground-moving target imaging and velocity estimation based on mismatched compression for bistatic forward-looking SAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 6, pp. 3277–3291, Jun. 2016, doi: 10.1109/TGRS.2016.2514494.

[2] W.-Q. Wang, "Forward-looking SAR imaging with frequency diverse array antenna," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 4191–4194, doi: 10.1109/IGARSS.2016.7730092.

[3] W. Pu et al., "Motion errors and compensation for bistatic forward-looking SAR with cubic-order processing," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 6940–6957, Dec. 2016, doi: 10.1109/TGRS.2016.2592536.

[4] J. Lu, L. Zhang, Y. Quan, Z. Meng, and Y. Cao, "Parametric azimuth-variant motion compensation for forward-looking multichannel SAR imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8521–8537, Oct. 2021, doi: 10.1109/TGRS.2020.3047449.

[5] J. Lu, L. Zhang, P. Xie, Z. Meng, and Y. Cao, "High-resolution imaging of multi-channel forward-looking synthetic aperture radar under curve trajectory," *IEEE Access*, vol. 7, pp. 51211–51221, 2019, doi: 10.1109/ACCESS.2019.2911554.

[6] J. Wu, J. Yang, Y. Huang, H. Yang, and H. Wang, "Bistatic forward-looking SAR: Theory and challenges," in *Proc. IEEE Radar Conf.*, May 2009, pp. 1–4, doi: 10.1109/RADAR.2009.4976959.

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[7] L. Huan, Z. Jian-xiong, and F. Qiang, "Modelling and simulation of a bistatic forward-looking SAR configuration," in *Proc. Int. Conf. Comput. Vis. Remote Sens.*, Dec. 2012, pp. 340–343, doi: 10.1109/CVRS.2012.6421286.

[8] C. Yang, J. Shen, Z. Jiang, C. She, and X. Zou, "A multi-channel radar forward-looking imaging algorithm based on super-resolution technique," in *Proc. China Int. SAR Symp. (CISS)*, Shanghai, China, Oct. 2018, pp. 1–4, doi: 10.1109/SARS.2018.8552029.

[9] Y. Chen, W. Li, R. Chen, J. Wu, and J. Yang, "Sidelobe suppression for multichannel forward-looking SAR imaging based on spatial smoothing coherence factor," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Pasadena, CA, USA, Jul. 2023, pp. 8296–8298, doi: 10.1109/igarss52108.2023.10281946.

[10] J. Wu, Y. Huang, J. Yang, W. Li, and H. Yang, "First result of bistatic forward-looking SAR with stationary transmitter," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Vancouver, BC, Canada, Jul. 2011, pp. 1223–1226, doi: 10.1109/IGARSS.2011.6049419.

[11] J. Wu, Z. Li, Y. Huang, J. Yang, H. Yang, and Q. H. Liu, "Focusing bistatic forward-looking SAR with stationary transmitter based on keystone transform and nonlinear chirp scaling," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 1, pp. 148–152, Jan. 2014, doi: 10.1109/LGRS.2013.2250904.

[12] Y. Li, T. Zhang, H. Mei, Y. Quan, and M. Xing, "Focusing translational-variant bistatic forward-looking SAR data using the modified Omega-K algorithm," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5203916, doi: 10.1109/TGRS.2021.3063780.

[13] H. Mei, Y. Li, M. Xing, Y. Quan, and C. Wu, "A frequency-domain imaging algorithm for translational variant bistatic forward-looking SAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1502–1515, Mar. 2020, doi: 10.1109/TGRS.2019.2943743.

[14] B. D. Rigling and R. L. Moses, "Polar format algorithm for bistatic SAR," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 40, no. 4, pp. 1147–1159, Oct. 2004, doi: 10.1109/TAES.2004.1386870.

[15] J. Xia, X. Lu, and W. Chen, "Multi-channel deconvolution for forward-looking phase array radar imaging," *Remote Sens.*, vol. 9, no. 7, p. 703, Jul. 2017, doi: 10.3390/rs9070703.

[16] W. Li, R. Chen, J. Yang, J. Wu, Y. Zhang, and Y. Huang, "A hybrid real/synthetic aperture scheme for multichannel radar forward-looking superresolution imaging," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023, doi: 10.1109/lgrs.2023.3309953.

[17] R. Chen, W. Li, K. Li, Y. Zhang, and J. Yang, "A super-resolution scheme for multichannel radar forward-looking imaging considering failure channels and motion error," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023, doi: 10.1109/LGRS.2023.3234264.

[18] J. Lu, L. Zhang, S. Wei, and Y. Li, "Resolution enhancement for forwarding looking multi-channel SAR imagery with exploiting space–time sparsity," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5200617, doi: 10.1109/tgrs.2022.3232392.

[19] X. Jiang, Y. Huang, J. Yang, and W. Li, "An improved monopulse forward-looking imaging algorithm," in *Communications, Signal Processing, and Systems* (Lecture Notes in Electrical Engineering), vol. 202, Q. Liang et al., Eds., New York, NY, USA: Springer, 2012, doi: 10.1007/978-1-4614-5803-6_17.

[20] C. Cheng, X.-D. Zhou, M. Gao, Z.-L. Zong, Y.-X. Ji, and B. Yu, "Research on monopulse forward-looking high-resolution imaging algorithm based on adaptive iteration," *Defence Technol.*, vol. 16, no. 1, pp. 158–171, Feb. 2020.

[21] X. Zhang, M. He, Z. He, H. Su, and J. Zhang, "Research on monopulse forward-looking imaging airborne radar system," in *Proc. IET Int. Radar Conf.*, Apr. 2013, pp. 1–3.

[22] Y. Zhang, D. Mao, Q. Zhang, Y. Zhang, Y. Huang, and J. Yang, "Airborne forward-looking radar super-resolution imaging using iterative adaptive approach," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 7, pp. 2044–2054, Jul. 2019, doi: 10.1109/JSTARS.2019.2920859.

[23] W. Li, M. Li, L. Zuo, H. Sun, H. Chen, and X. Lu, "Azimuth super-resolution of forward-looking imaging based on Bayesian learning in complex scene," *Signal Process.*, vol. 187, Oct. 2021, Art. no. 108141.

[24] G. Nwalozie, "A simple comparative evaluation of adaptive beam forming algorithms," *Int. J. Eng. Innov. Technol.*, vol. 2, no. 7, pp. 1–8, 2013.

[25] J. Lu, L. Zhang, and Z. Meng, "A pulse-by-pulse Doppler ambiguity resolving algorithm for FLMC-SAR imaging based on fast factorized back-projection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5205419, doi: 10.1109/TGRS.2024.3351766.

[26] J. Lu, X. Wang, Y. Cao, and L. Zhang, "Sparsity-based joint array calibration and ambiguity resolving for forward-looking multi-channel SAR imagery," *Remote Sens.*, vol. 15, no. 3, p. 647, Jan. 2023, doi: 10.3390/rs15030647.

[27] R. Bamler, "A comparison of range-Doppler and wavenumber domain SAR focusing algorithms," *IEEE Trans. Geosci. Remote Sens.*, vol. 30, no. 4, pp. 706–713, Jul. 1992, doi: 10.1109/36.158864.

[28] Z. Sun, J. Wu, Z. Li, Y. Huang, and J. Yang, "Highly squint SAR data focusing based on keystone transform and azimuth extended nonlinear chirp scaling," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 1, pp. 145–149, Jan. 2015, doi: 10.1109/LGRS.2014.2329554.

[29] N. B. Karahanoglu and H. Erdogan, "Forward-backward search for compressed sensing signal recovery," in *Proc. 20th Eur. Signal Process. Conf. (EUSIPCO)*, Bucharest, Romania, Aug. 2012, pp. 1426–1433.

[30] S. Shao, L. Zhang, H. Liu, and Y. Zhou, "Accelerated translational motion compensation with contrast maximisation optimisation algorithm for inverse synthetic aperture radar imaging," *IET Radar, Sonar Navigat.*, vol. 13, no. 2, pp. 316–325, Feb. 2019.

[31] A. Evers and J. A. Jackson, "A generalized phase gradient autofocus algorithm," *IEEE Trans. Comput. Imag.*, vol. 5, no. 4, pp. 606–619, Dec. 2019, doi: 10.1109/TCI.2019.2899453.



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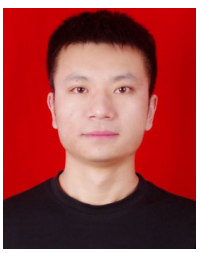
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